IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Instructions and guidelines (read carefully)

Instructions

1. Insert your name as it appears on your Online Campus learner profile in the file name. Save the file as: **First Name Last Name M10U4 Capstone Project** – **e.g., Lilly Smith M10U4 Capstone Project.**
2. Write all your answers and upload your screenshots in this PowerPoint presentation in the slides provided.
3. Convert your final presentation into **.pdf** format and submit this **PDF document**. No other file types will be accepted.

Note: Please ensure that you have checked your course calendar for the due date for this assignment.

Guidelines

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1. Complete the slides with relevant details (you are free to add additional slides, charts, and tables).
2. Add any additional unused charts or tables to the appendix that you think are relevant to your analysis.

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Outline

- Executive summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Reference: https://github.com/KarinaMurugan/Karina-Murugan-IBM-Project/tree/main

Executive summary

Predicting Falcon 9 Landing Success
A Machine Learning Approach to Cost Advantage

Objective

- Assess whether Falcon 9 first-stage landing success can be predicted using historical data

Methodology

- Data integration: SpaceX API + Wikipedia
- Data preparation: cleaning, imputation, feature engineering
- Exploratory Data Analysis (EDA) + SQL for pattern identification
- Interactive analytics: Folium maps & Plotly Dash dashboard
- Machine learning: Logistic Regression, SVM, Decision Tree, KNN
- Model evaluation using accuracy and ROC-AUC

Key Results

- All models achieved ~83% accuracy
- **SVM performed best** (ROC-AUC: 0.96)
- Landing success influenced by:
  - Launch site
  - Orbit type
  - Time (learning effect)

Insight

- Landing outcomes are **predictable**, not random
- Enables improved cost forecasting and operational decision-making

Introduction

Project Background & Context

- SpaceX has disrupted the aerospace industry through reusable rocket technology
- Falcon 9's first-stage recovery significantly reduces launch costs
- Landing success is critical to achieving these cost efficiencies

Problem Statement

- Landing outcomes are uncertain and influenced by multiple factors
- Limited transparency makes it difficult to:
  - Estimate true launch costs
  - Optimize mission planning
  - Assess competitive positioning

Project Objective

- Develop a predictive model to determine the likelihood of landing success
- Identify key factors driving successful landings

Section 1

Methodology

Methodology

Executive summary

- Data collection methodology
- Perform data wrangling.
- Perform exploratory data analysis (EDA) using visualization and SQL.
- Perform interactive visual analytics using Folium and Plotly Dash.
- Perform predictive analysis using classification models.

Data collection

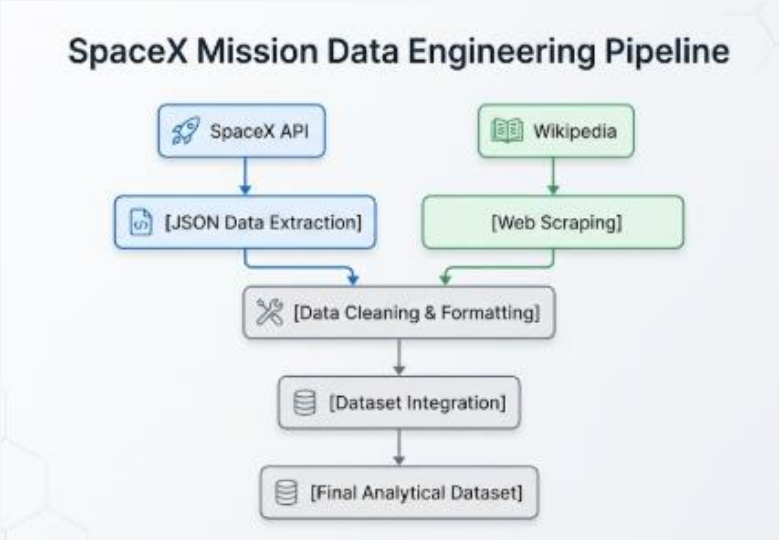
Data Sources

- **SpaceX API**
  - Launch data
  - Rocket and booster details
  - Payload information
  - Launch site data

- **Wikipedia (Web Scraping)**
  - Historical launch records
  - Booster reuse information
  - Additional mission attributes

Data Collection Process (Key Steps)

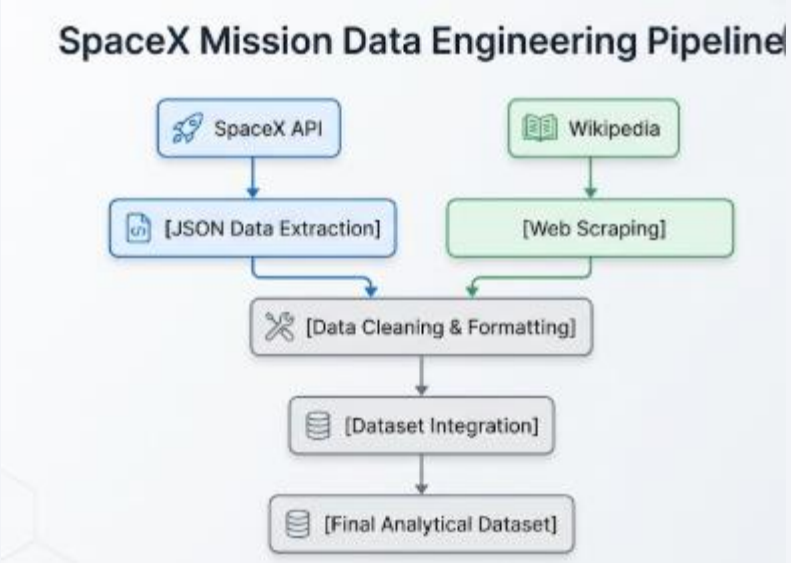
- Extract structured JSON data from API endpoints
- Parse and normalize nested data into tabular format
- Scrape relevant tables from Wikipedia
- Clean and standardize scraped data
- Merge datasets using common identifiers (e.g., flight number, booster ID)



Data collection – SpaceX REST API Integration

Technical Methodology

- **Extraction:** Utilized the SpaceX v4 API to retrieve historical launch mission data.
- **Data Cleaning:** Parsed JSON responses using pandas.json_normalize.
  - Handled missing values in Payload Mass via mean imputation to prevent model bias.
- **Feature Engineering:** Engineered a binary target variable (Class) where 1 indicates a successful landing and 0 indicates a failure.



Outcome: Successfully generated a structured dataset with 200+ rows, serving as the foundation for all subsequent predictive modeling.

API Strategy	Data Cleaning	Data Integrity
Used requests library to fetch JSON from v4 endpoints.	Flattened nested JSON with json_normalize.	Replaced null Payload Mass with feature mean.

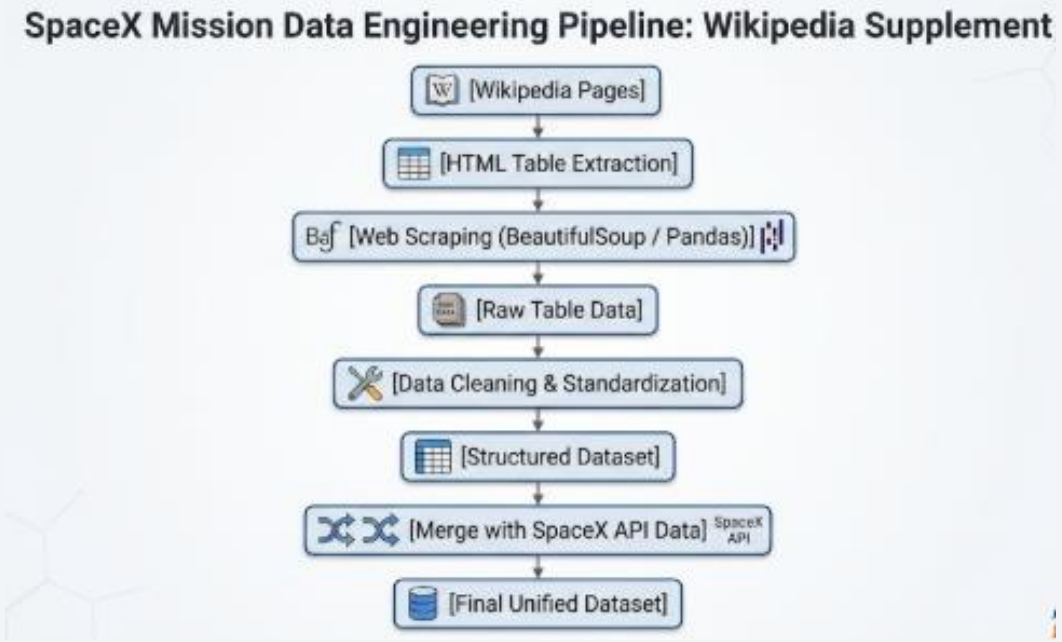
Data collection – Scraping

Objective

- Extract supplementary historical launch and booster data from Wikipedia to enrich SpaceX API dataset
- Improve dataset completeness and feature depth for analysis and modelling

Web Scraping Process (Key Steps)

- Identified relevant Wikipedia pages (Falcon 9 & Falcon Heavy launch history)
- Inspected HTML structure to locate structured tables
- Extracted tables using Python web scraping libraries (e.g., BeautifulSoup / Pandas read_html)
- Parsed and cleaned raw HTML table data
- Standardized column names and data formats
- Removed inconsistencies and irrelevant rows
- Converted scraped data into structured tabular format
- Merged with SpaceX API dataset using common identifiers



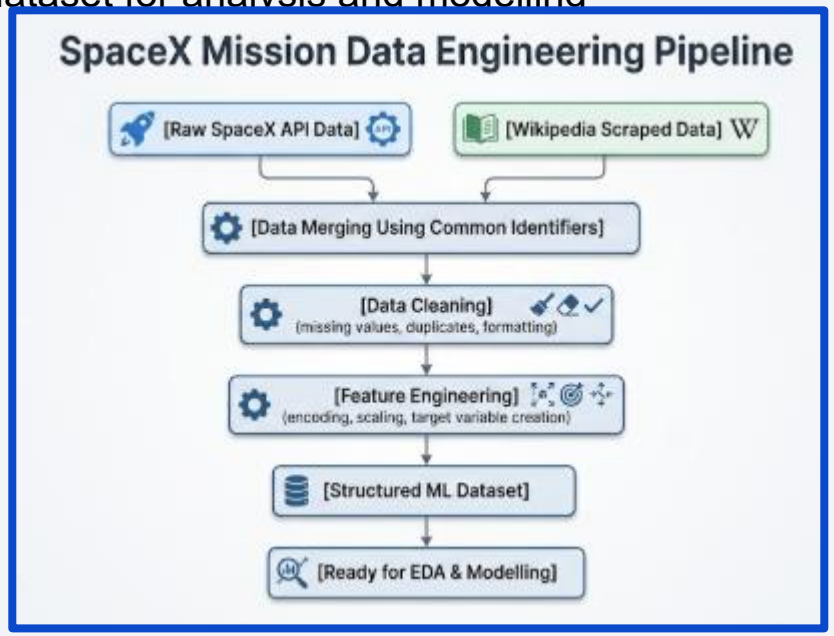
Challenges Addressed

- Inconsistent table formatting across Wikipedia pages
- Missing or partially structured values
- Data standardization across multiple sources

Data wrangling

Objective

- Transform raw SpaceX API and Wikipedia data into a clean dataset for analysis and modelling



Key Processing Steps

- Merged SpaceX API and Wikipedia datasets using common identifiers
- Standardised column names and data formats across sources
- Handled missing values (mean imputation for numerical features)
- Removed duplicates and irrelevant records
- Encoded categorical variables for machine learning
- Scaled numerical features for model consistency
- Created target variable:
  - 1 = successful landing
  - 0 = failed landing

Output

- Clean, structured dataset ready for EDA and machine learning
- Consistent feature set across all observations
- Fully model-ready data pipeline

EDA with data visualization

Key Visualisations & Rationale

- **Bar /Donut Charts (Success Rate by Category)**
  - Used to compare landing success across launch sites and orbit types
  - Helps identify operational and environmental differences in performance
- **Scatter Plots (Payload vs Success Outcome)**
  - Used to assess relationship between payload mass and landing success
  - Useful for identifying trends or thresholds affecting outcomes
- **Time Series Analysis (Success over Time)**
  - Used to track performance improvement from 2010–2020
  - Highlights learning effects and technological advancement
- **Geospatial Mapping (Folium Maps)**
  - Used to visualise launch site locations globally
  - Provides spatial context to operational performance
- **Interactive Dashboard (Plotly Dash)**
  - Used for dynamic filtering of launch site and payload variables
  - Enables deeper exploratory insights beyond static charts

Key Insights from EDA

- Significant variation in success rates across launch sites
- Clear improvement in landing success over time
- Orbit type and payload range influence landing probability
- KSC LC-39A shows consistently higher performance

EDA with SQL

Objective

- Use SQL queries to extract structured insights from the SpaceX dataset and validate patterns observed during EDA

Summary of SQL Queries Performed

- Identified unique launch sites in the dataset
- Extracted launch records for KSC sites (pattern-based filtering using “KSC%”)
- Calculated total payload mass carried by different booster categories
- Computed average payload mass for specific booster versions (e.g. F9 v1.1)
- Retrieved date of first successful drone ship landing
- Counted total successful vs failed landing outcomes across all missions
- Identified boosters associated with maximum payload mass missions
- Analysed landing outcomes by month, booster version and launch site (2017 focus)
- Ranked landing outcomes (success/failure types) over a defined time range (2010–2017)

Key Outcome

- SQL enabled structured filtering, aggregation, and temporal analysis of landing outcomes across multiple dimensions (site, booster type, payload, and time)

Build an interactive map with Folium

Objective

- Visualise SpaceX launch sites geographically and analyse spatial relationships affecting launch operations and outcomes

Summary of Map Objects Created

- **Launch Site Markers (All Sites)**
  - Plotted each launch site using interactive markers
  - Used to identify exact geographic locations of all SpaceX launch facilities
- **Circles around Launch Sites**
  - Added circular overlays to represent launch site coverage areas
  - Helps visualise operational range and spatial influence zones
- **Colour-Coded Markers (Success vs Failure)**
  - Green markers = successful landings
  - Red markers = failed landings
  - Enables immediate visual comparison of performance by location
- **Mouse Position Coordinates Tool**
  - Displays latitude and longitude dynamically
  - Used to explore spatial relationships and surrounding infrastructure
- **Distance Lines to Nearby Infrastructure**
  - Lines drawn to nearest coastline, railway, highway, and city
  - Includes distance markers for each reference point

Purpose of Map Features

- To understand how geographic location influences launch and landing success\
- To assess proximity to infrastructure (coastlines, transport routes, urban areas)
- To support evaluation of operational safety, logistics efficiency, and site suitability
- To provide spatial context for comparing performance across launch sites

Build a dashboard with Plotly Dash

Objective

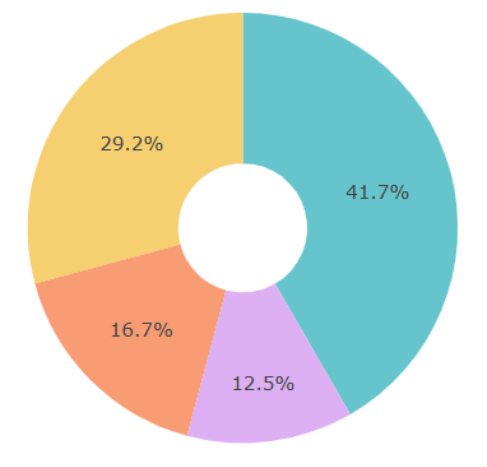
- Develop an interactive dashboard to explore Falcon 9 landing success patterns across key variables

Dashboard Components & Interactions

- **Launch Site Filter (Dropdown Selection)**
  - Allows users to view all sites or isolate a single launch site
  - Enables comparison between overall performance and site-specific outcomes
- **Payload Mass Range Slider**
  - Interactive filter to select payload intervals
  - Used to analyse how payload size influences landing success probability
- **Success Rate Donut Chart**
  - Displays proportion of successful vs failed landings
  - Updates dynamically based on selected launch site
- **Scatter Plot (Payload Mass vs Success Outcome)**
  - Visualises relationship between payload weight and landing success
  - Includes differentiation by booster version

Purpose of Dashboard Design

- Enable interactive exploration of landing success drivers
- Support dynamic comparison across launch sites and payload ranges
- Provide both high-level summaries and detailed drill-down analysis
- Translate static EDA findings into a user-driven analytical tool



Predictive analysis (classification)

Objective

- Develop and evaluate classification models to predict Falcon 9 landing success
- Identify the best-performing model based on predictive performance

Model Development Process

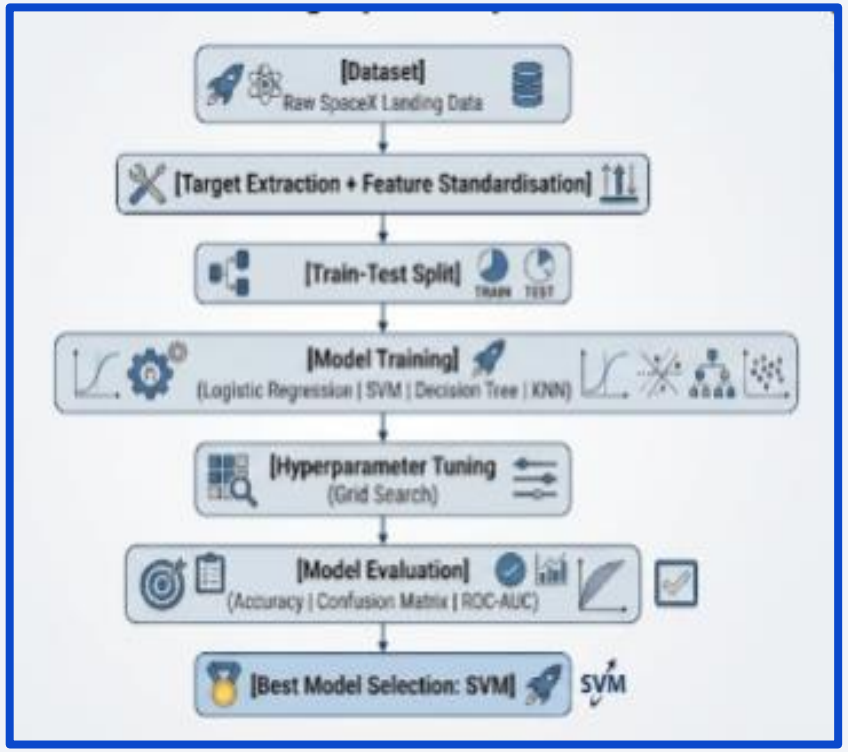
- Extracted target variable (landing success: 1 = success, 0 = failure)
- Standardised feature variables to ensure model consistency
- Split dataset into training and testing sets
- Trained multiple classification models:
  - Logistic Regression
  - Support Vector Machine (SVM)
  - Decision Tree
  - K-Nearest Neighbors (KNN)
- Performed hyperparameter tuning using Grid Search
- Evaluated models using:
  - Accuracy
  - Confusion Matrix
  - ROC-AUC Score

Model Selection Criteria

- Primary focus: ROC-AUC (model discrimination ability)
- Secondary: Accuracy and generalisation performance

Outcome

- SVM identified as the best-performing model
- Achieved highest ROC-AUC score: **0.9583**
- Demonstrated strongest ability to distinguish between successful and failed landings



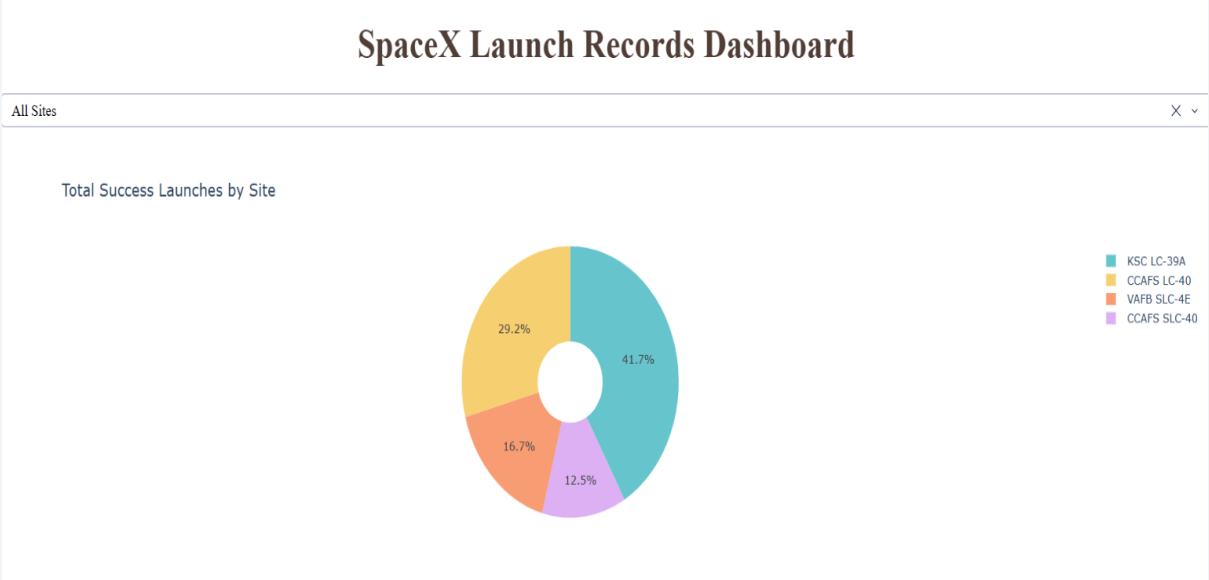
Results

1. Exploratory Data Analysis (EDA) Results

- Launch success rates vary significantly by launch site
  - KSC LC-39A recorded the highest success rate (~76.9%)
- Orbit type strongly influences landing success, indicating varying mission complexity
- Clear upward trend in success rates over time (2010–2020)
- Payload mass shows no simple linear relationship, suggesting complex interactions

2. Interactive Analytics (Dashboard Demo)

- Dashboard enables dynamic filtering by:
  - Launch site
  - Payload mass range
- Pie chart visualises success vs failure distribution per site
- Scatter plot reveals relationship between payload mass and landing outcome
- Interactive tools allow real-time exploration of performance patterns



3. Predictive Analysis Results

- All models achieved similar baseline accuracy (~83%)
- SVM outperformed all models:
  - ROC-AUC: 0.9583
  - Strongest classification performance
- Indicates landing success is driven by non-linear relationships between features
- Model successfully distinguishes between successful and failed landings with high reliability

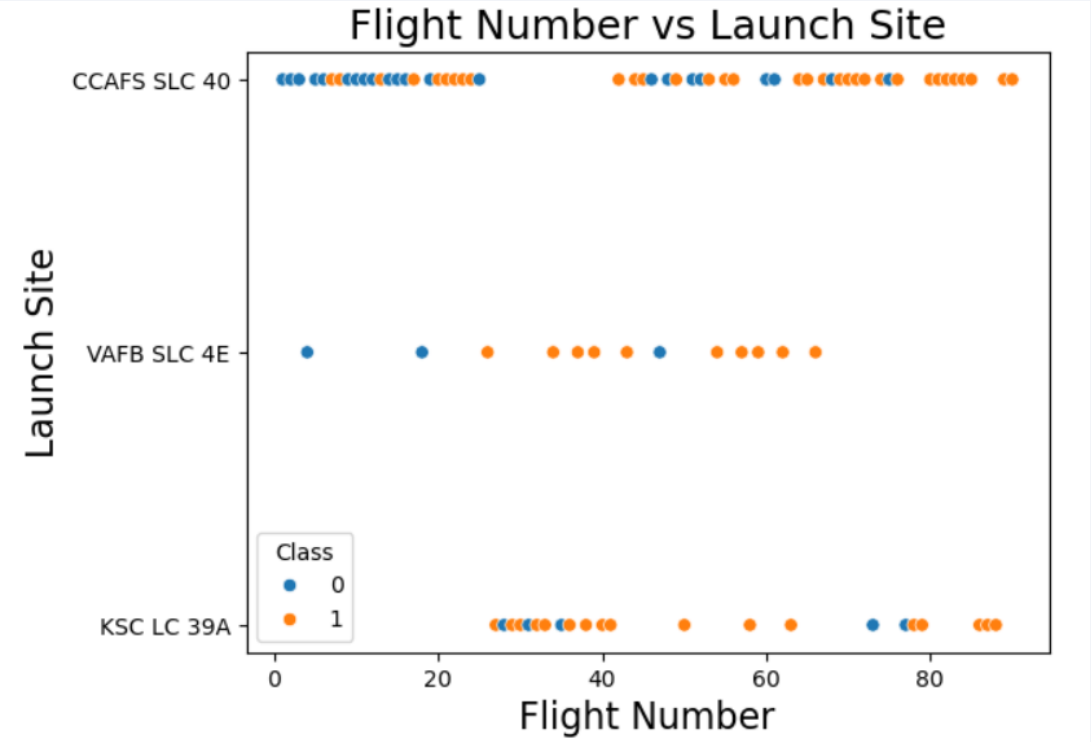
Section 2

Insights drawn from EDA

Flight number vs. launch site

Observation from Scatter Plot

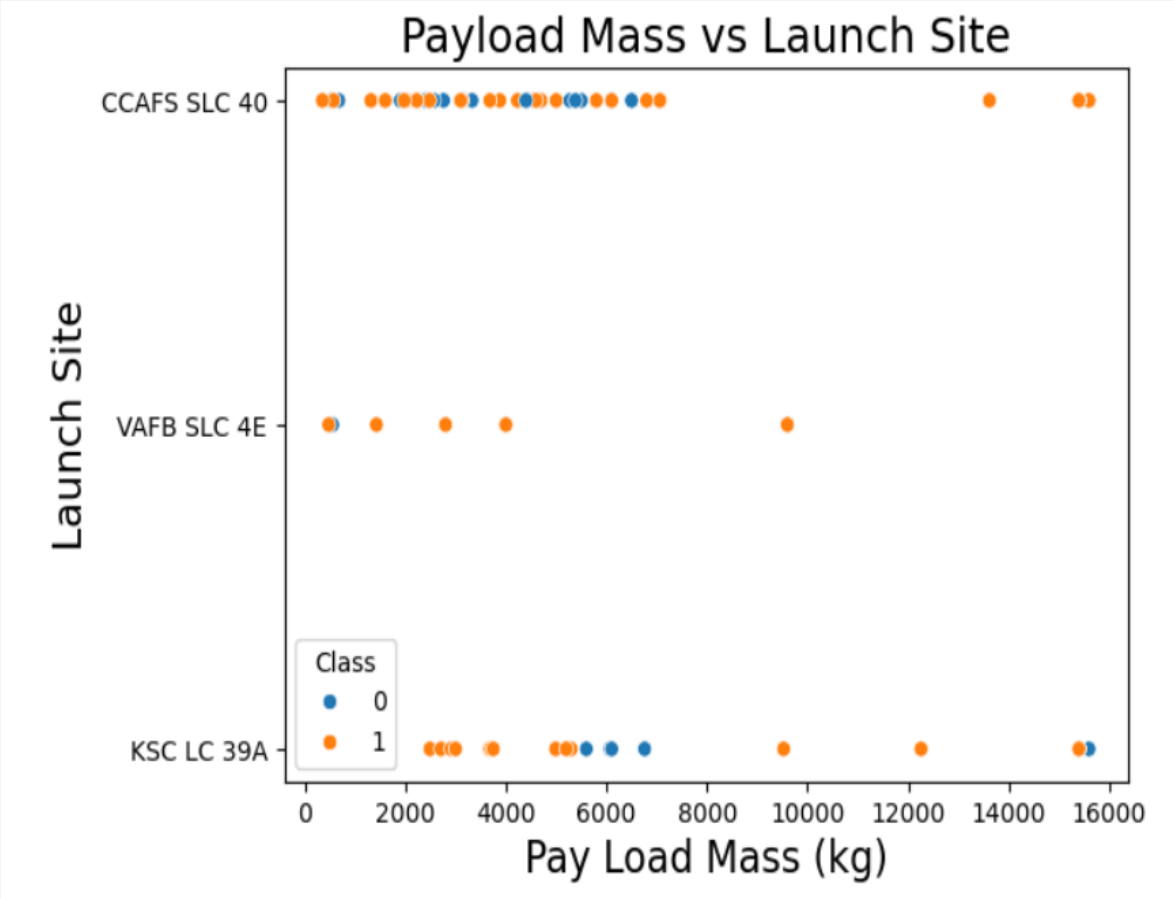
- Early flight numbers are concentrated at a limited number of launch sites
- Over time, launch activity expands across multiple sites
- Certain launch sites (e.g. KSC LC-39A) appear more frequently in later flights
- Suggests progressive scaling and diversification of launch infrastructure



Payload vs. launch site

Observation from Scatter Plot

- Payload masses vary significantly across different launch sites
- Certain launch sites handle a wider range of payloads, indicating higher operational flexibility
- Heavier payloads are more frequently associated with specific sites (e.g. KSC LC-39A)
- Some sites show a concentration of lighter payload missions



Key Insight

- Payload distribution is not uniform across launch sites
- Launch site selection is influenced by mission requirements and payload characteristics
- High-capacity sites are more frequently used for heavier and more complex missions

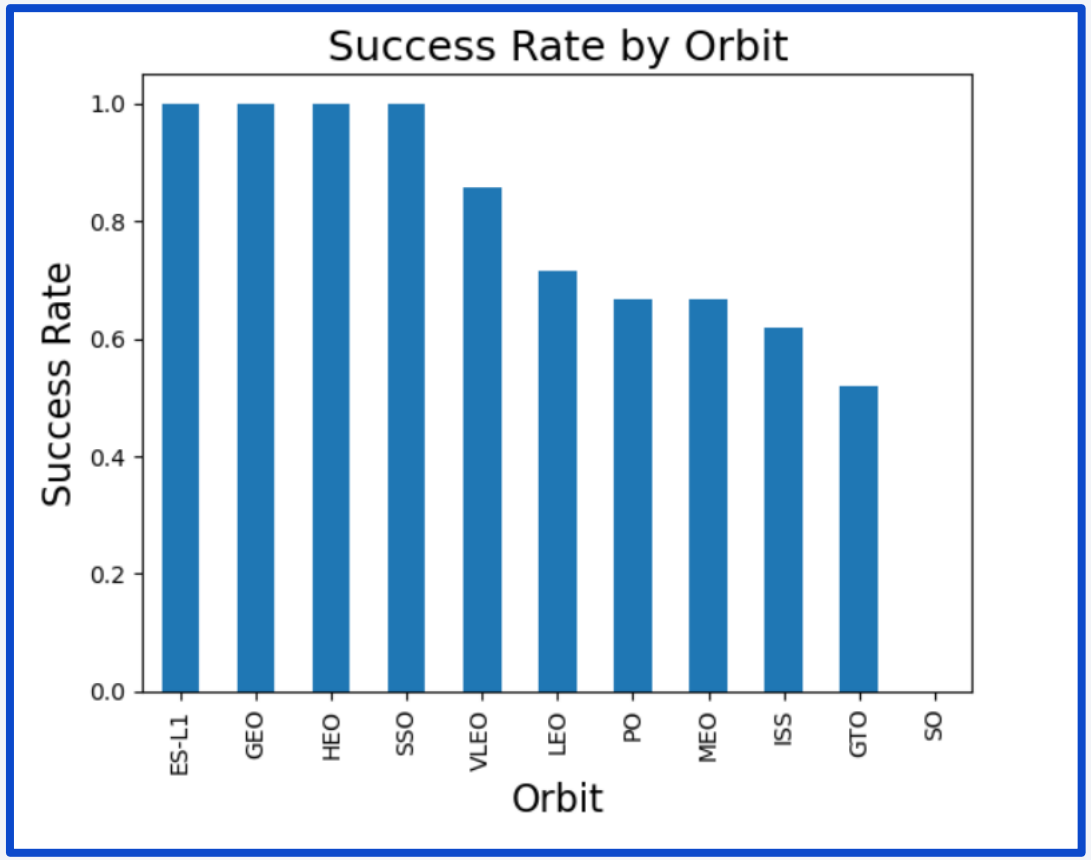
Success rate vs. orbit type

Observation from Bar Chart

- Landing success rates vary significantly across orbit types
- Certain orbit categories demonstrate consistently higher success rates
- Some orbit types show noticeably lower success performance, indicating higher mission difficulty
- Distribution of success is uneven across orbital classifications

Key Insight

- Orbit type is a strong determinant of landing success rate
- Different orbital missions carry varying levels of technical complexity and operational risk
- Higher success rates are concentrated in a subset of orbit categories



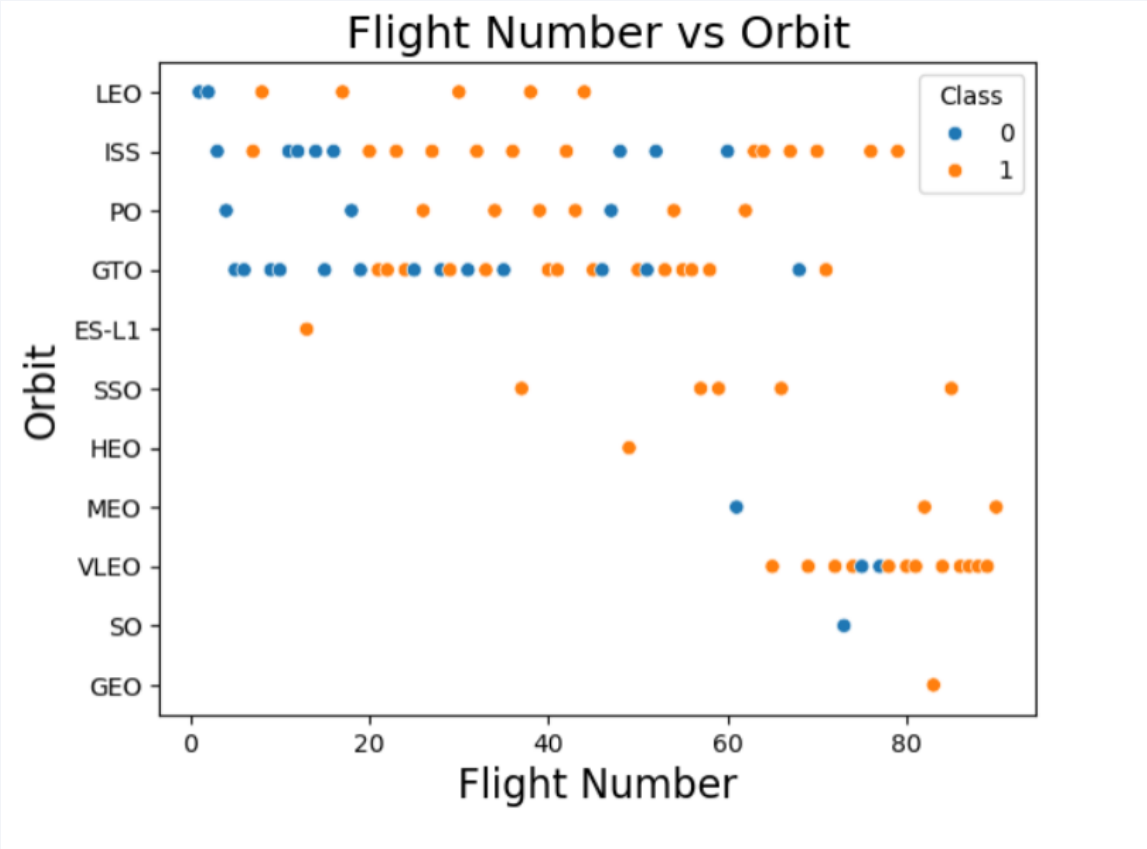
Flight number vs. orbit type

Observation from Scatter Plot

- Early flight numbers are concentrated in a limited set of orbit types
- As flight number increases, missions are distributed across a wider variety of orbit types
- Certain orbit types appear more frequently in later missions, indicating increased usage
- Higher flight numbers show greater diversity in mission-orbit combinations

Key Insight

- There is a clear relationship between mission progression and orbit complexity
- Over time, SpaceX expands from a limited set of orbit types to a broader operational range
- Suggests increasing technical capability and mission diversification



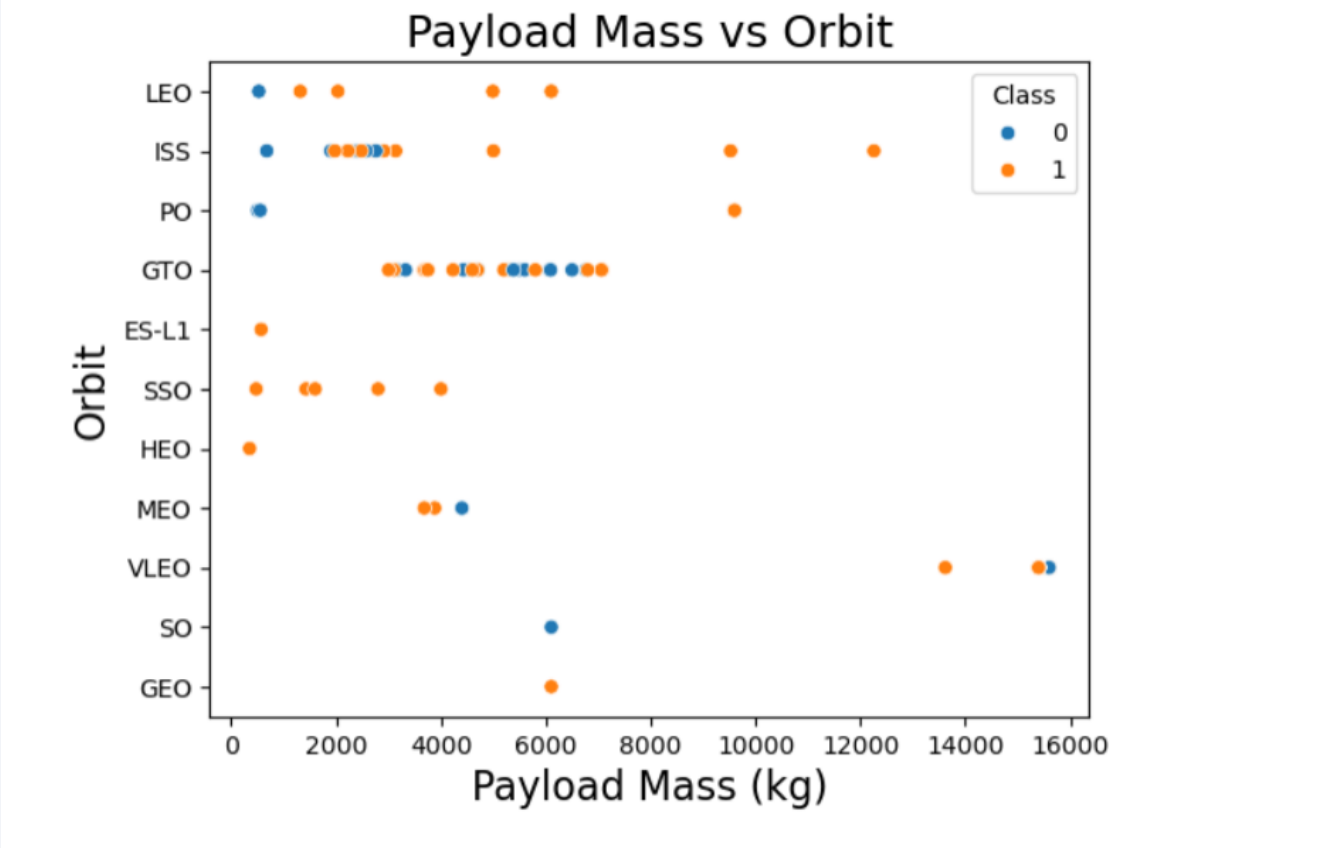
Payload vs. orbit type

Observation from Scatter Plot

- Payload masses vary significantly across different orbit types
- Some orbit types are associated with a wider range of payload values
- Certain orbits are predominantly used for lighter payload missions
- Heavier payloads appear more concentrated in specific orbit categories

Key Insight

- Payload capacity is closely linked to orbit type selection
- Different orbital missions impose varying technical and structural constraints on payload size
- Some orbit types support more high-mass, complex missions, while others are limited to lighter payloads



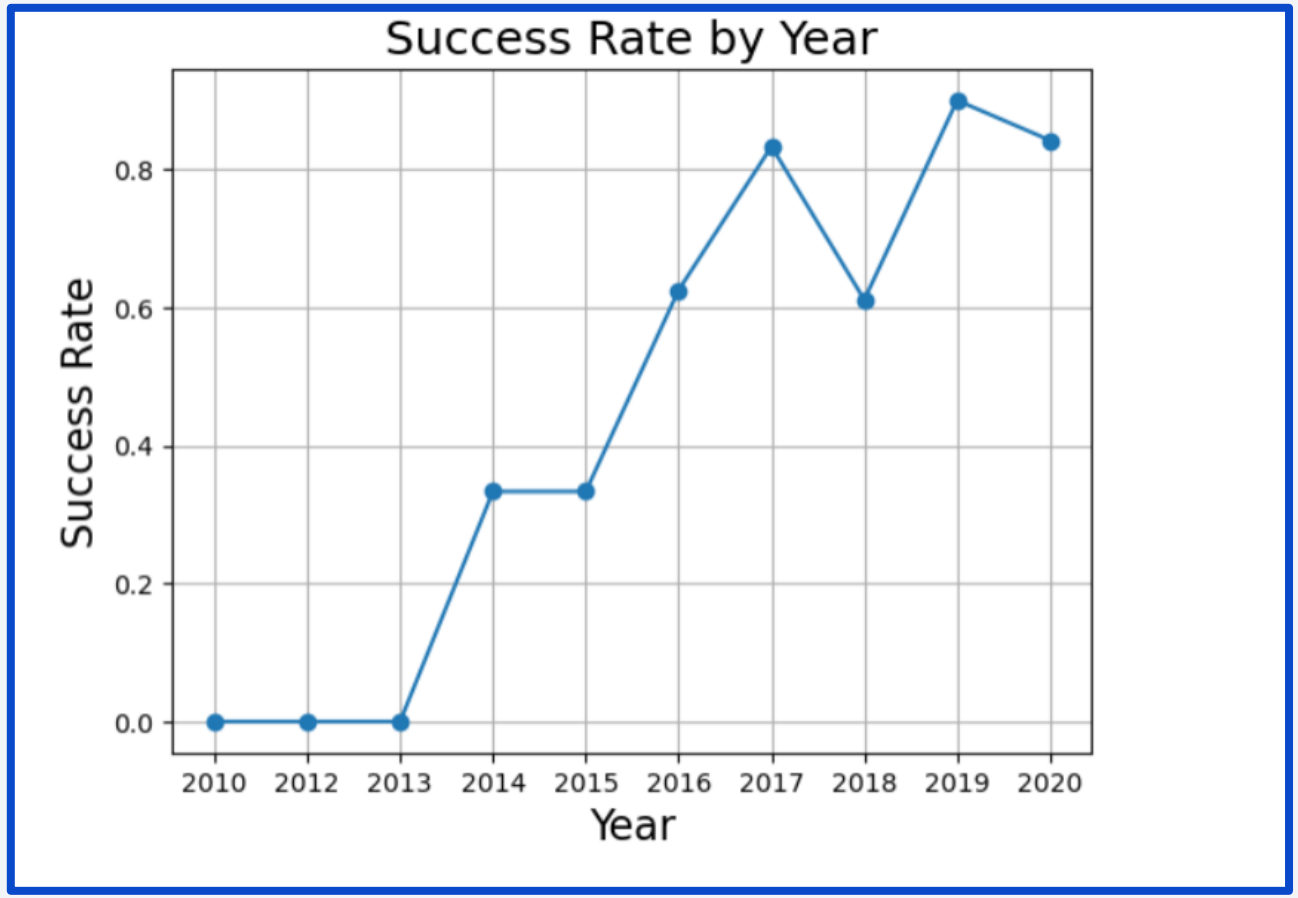
Launch success yearly trend

Observation from Line Chart

- Launch success rate shows a clear upward trend over time
- Early years exhibit lower and more variable success rates
- From mid-period onwards, success rates become more stable and consistently high
- Recent years show the highest overall success performance

Key Insight

- There is a strong positive trend in landing success over time
- Indicates continuous technological improvement and operational learning
- Suggests that experience significantly contributes to mission success



All launch site names

SQL Query Objective

- Retrieve the list of all distinct SpaceX launch sites in the dataset

Query Result

- CCAFS SLC-40
- CCAFS LC-40
- KSC LC-39A
- VAFB SLC-4E

Explanation

- The query identifies all unique launch locations used by SpaceX across recorded missions
- This helps establish the geographic distribution of launch operations
- These sites form the basis for further analysis of site-based performance differences

Launch site names begin with “KSC”

- The query uses a pattern-matching filter (LIKE operator) to isolate launch records from Kennedy Space Center sites
- This helps focus analysis on a specific geographic launch location
- These records are used to compare performance patterns for KSC-based launches

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2017-03-16	6:00:00	F9 FT B1030	KSC LC-39A	EchoStar 23	5600	GTO	EchoStar	Success	No attempt
2017-03-30	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)
2017-05-01	11:15:00	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO	Success	Success (ground pad)
2017-05-15	23:21:00	F9 FT B1034	KSC LC-39A	Inmarsat-5 F4	6070	GTO	Inmarsat	Success	No attempt

Total payload mass

SQL Query Objective

- Calculate the total payload mass carried by boosters associated with NASA missions

Query Result

- Total payload carried by NASA boosters:

total_payload
45596

Explanation

- The query aggregates payload mass using a SUM function for records linked to NASA-related missions
- This provides insight into the overall mission load carried by NASA-associated boosters
- Useful for understanding mission scale and contribution to total launch activity

Average payload mass by F9 v1.1

SQL Query Objective

- Calculate the average payload mass carried by the Falcon 9 v1.1 booster version

Query Result

- Average payload mass (F9 v1.1):

avg_payload
2928.4

Explanation

- The query uses an AVG aggregation function to compute the mean payload mass for all missions using the F9 v1.1 booster version
- This helps assess the typical mission load capacity for this specific booster generation
- Useful for comparing performance differences across booster versions

First successful ground landing date

SQL Query Objective

- Retrieve the **earliest date of a successful landing on a drone ship**

Query Result

- First successful drone ship landing date:

min(Date)
2016-04-08

Explanation

- The earliest record represents the first successful recovery at sea using a drone ship
- This marks a key milestone in SpaceX's development of reusable rocket technology
- The query filters records for successful drone ship landings and sorts results by date in ascending order

Successful drone ship landing with payload between 4,000 and 6,000

SQL Query Objective

- Retrieve the names of boosters that successfully landed on a drone ship
- Filter results where payload mass is between 4,000 and 6,000 kg

Query Result

- Booster names:

Booster_Version
F9 FT B1032.1
F9 B4 B1040.1
F9 B4 B1043.1

Explanation

- The query applies multiple filters:
  a. Landing outcome = successful drone ship landing
  b. Payload mass range = 4,000–6,000 kg
- This isolates missions with moderate payload demand under successful sea recovery conditions
- Helps identify which boosters are most reliable under mid-range operational loads

Total number of successful and failed mission outcomes

SQL Query Objective

- Calculate the total number of successful and failed mission outcomes in the dataset

Query Result

- Successful missions:
- Failed missions:

mission_outcome	num_missions
Success	100
Failure	1

Explanation

- The query groups mission outcomes and counts occurrences of each category
- This provides a high-level summary of overall mission performance distribution
- It helps assess the reliability and success rate of Falcon 9 missions

Boosters carried maximum payload

SQL Query Objective

- Identify the booster(s) associated with the maximum payload mass in the dataset

Query Result

- Booster name(s):

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Short Explanation

- The query uses an aggregation function to determine the maximum payload value
- It then filters the dataset to return the booster(s) linked to this highest payload mission
- This highlights which boosters were used for the most demanding (heaviest) launch conditions

2015 launch records

SQL Query Objective

- Retrieve records showing:
  - Month names
  - Booster versions
  - Launch sites
  - Successful ground pad landing outcomes
- Filtered for missions in the year 2017

Explanation

- The query uses an aggregation function to determine the maximum payload value
- It then filters the dataset to return the booster(s) linked to this highest payload mission
- This highlights which boosters were used for the most demanding (heaviest) launch conditions

Query Result

- Records include:

Month_Name	Booster_Version	Launch_Site	Landing_Outcome
February	F9 FT B1031.1	KSC LC-39A	Success (ground pad)
May	F9 FT B1032.1	KSC LC-39A	Success (ground pad)
June	F9 FT B1035.1	KSC LC-39A	Success (ground pad)
August	F9 B4 B1039.1	KSC LC-39A	Success (ground pad)
September	F9 B4 B1040.1	KSC LC-39A	Success (ground pad)
December	F9 FT B1035.2	CCAFS SLC-40	Success (ground pad)

Rank landing outcomes between 2010-06-04 and 2017-03-20

SQL Query Objective

- Rank and count landing outcomes (e.g. *Success (ground pad), Failure (drone ship)*)
- Filter results for the period 2010-06-04 to 2017-03-20
- Display results in descending order of frequency

Query Result

- Landing outcomes ranked by frequency:

Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Explanation

- The query groups landing outcomes within the
- selected time period and counts their occurrences
- Results are sorted in descending order to highlight the most frequent outcomes first
- This provides a clear view of the dominant landing success and failure types during early mission years

Section 3

Launch Sites Proximities Analysis

Global Distribution of SpaceX Launch Sites (Folium Map)

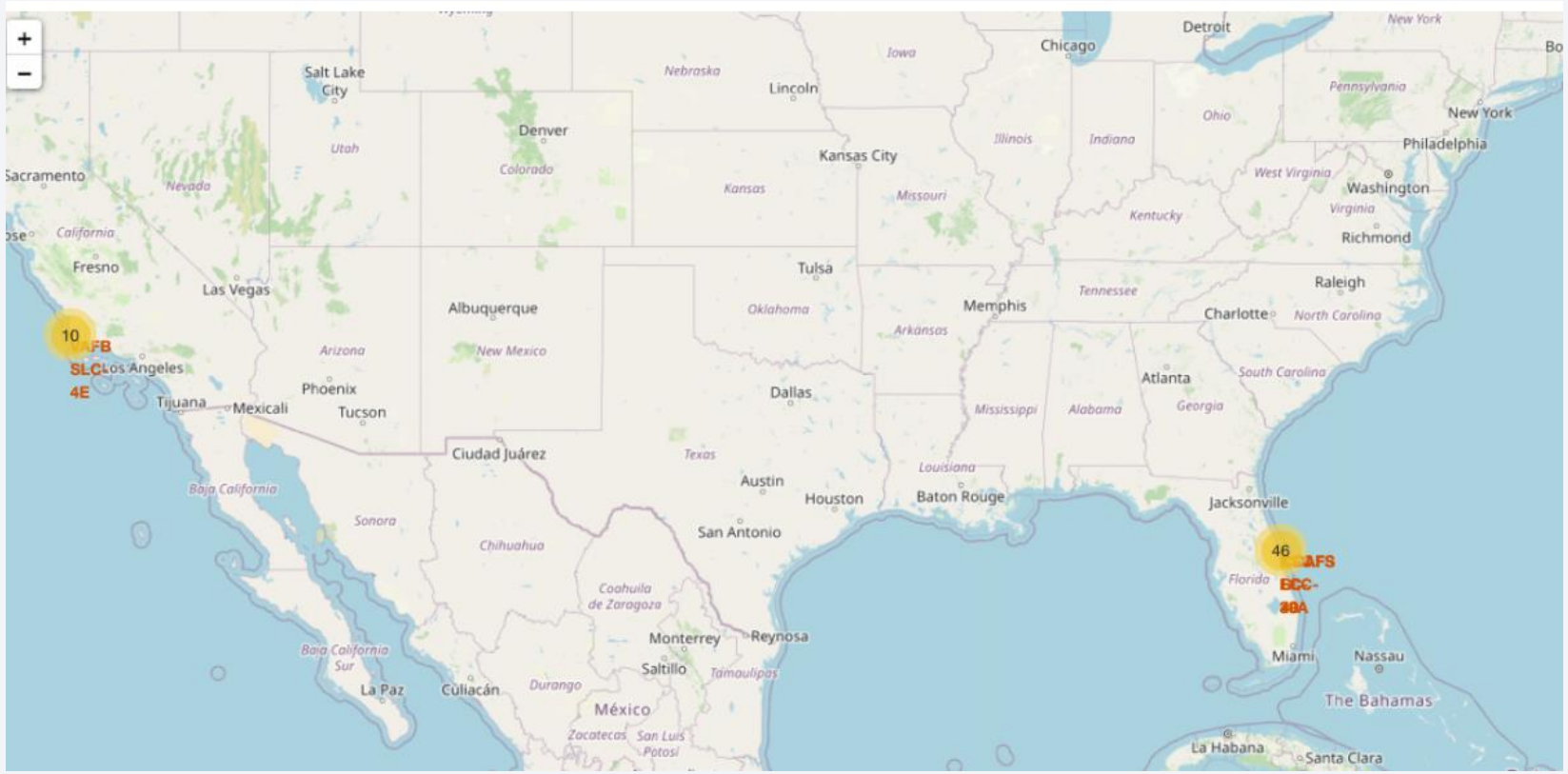
Findings:

Marker Icons: Clearly labeled icons identify specific launch sites (e.g., KSC LC-39A, SLC-40, VAFB SLC-4E).

Launch Sites: 4 launch sites where used

Proximity Vectors: The map clearly visualizes that all sites are located near coastal areas.

Interactive: The markers are interactive, allowing users to drill down into mission success rates at each specific site.



Geospatial Visualization of Falcon 9 Mission Outcomes by Launch Site

Findings:

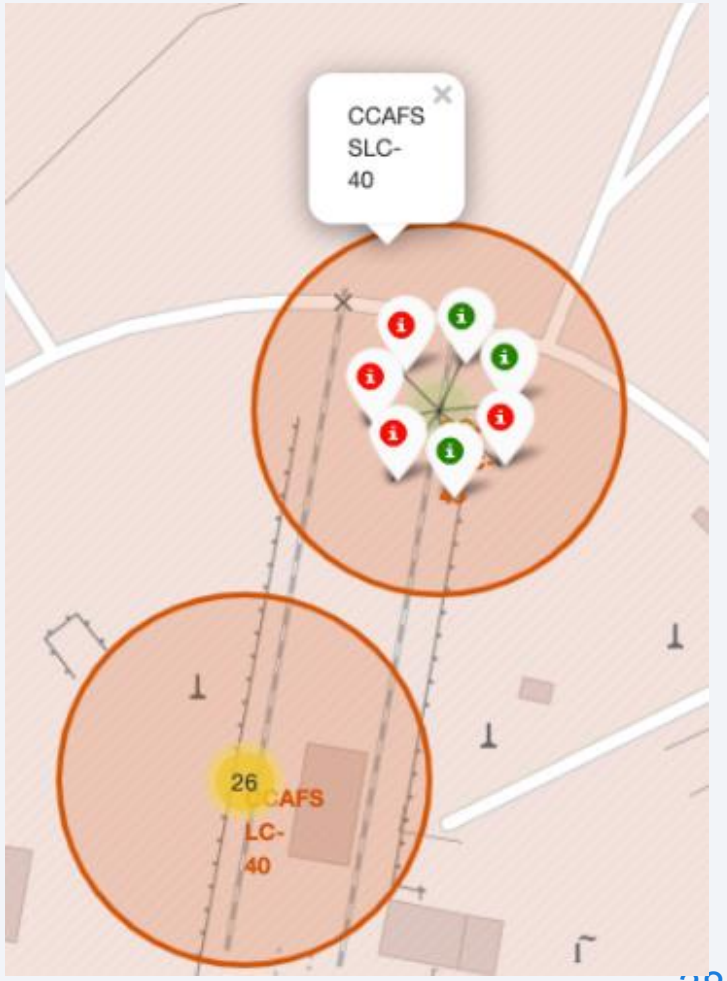
- 1. Visual Legend

  Color Coding:

  Green Markers: Successful Landing (Class = 1)

  Red Markers: Failed Landing (Class = 0)

- Geographic Context: Clearly label the launch sites



Key Insights/Findings

- Operational Reliability: Visualization highlights a higher density of green markers at mature launch sites (like KSC LC-39A), suggesting a correlation between site usage frequency and technical success.
- Failure Analysis: Failed landings (red markers) are largely concentrated in earlier project phases, visually confirming the iterative nature of the Falcon 9 engineering program.
- Regional Trends: The map demonstrates that success is geographically distributed, proving that the Falcon 9 landing system is robust across varying mission profiles and launch environments.

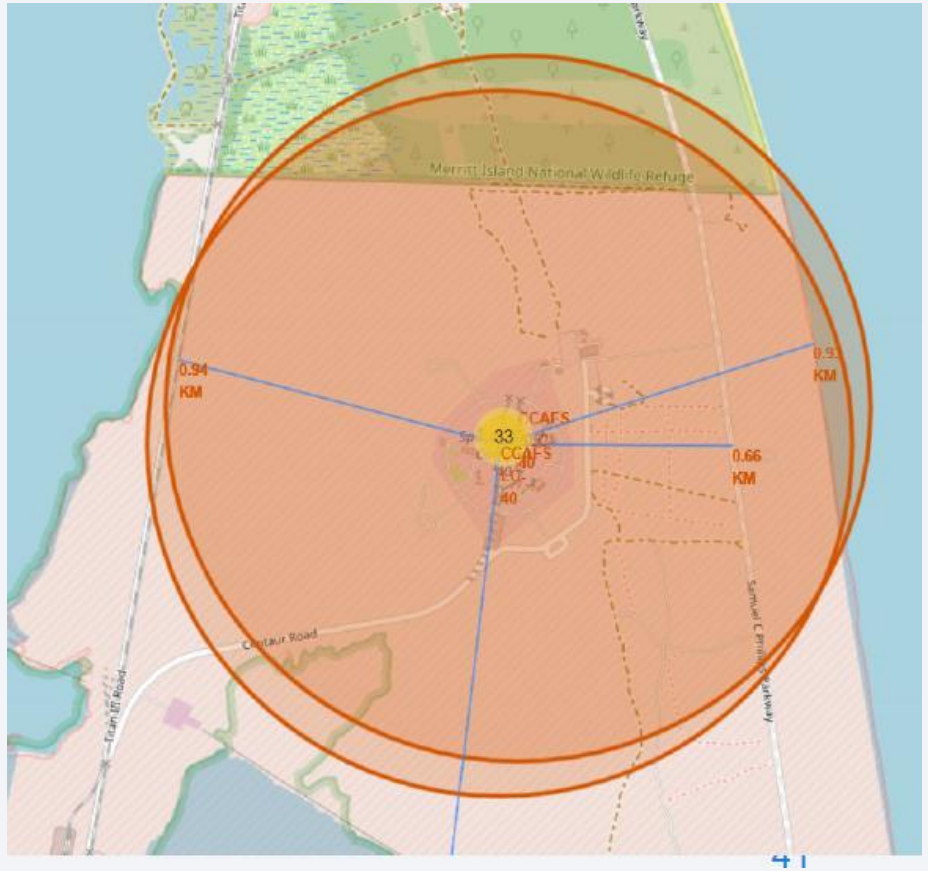
CAFS LC-40 with varying proximities

Key Points

- **Infrastructure Connectivity:** Calculated proximity to **major railways and highways** confirms efficient transport pipelines for heavy-lift rocket components, minimizing "stack-to-pad" lead times.

- **Safety & Compliance:** Strategic location relative to the **coastline** ensures a clear "safety corridor" for launches, strictly adhering to regulatory flight-path safety standards.

- **The Logistical "Goldilocks Zone":** The sites are optimized to balance two competing needs: proximity to major transport hubs (for cost reduction) and immediate access to the ocean (for safety compliance).



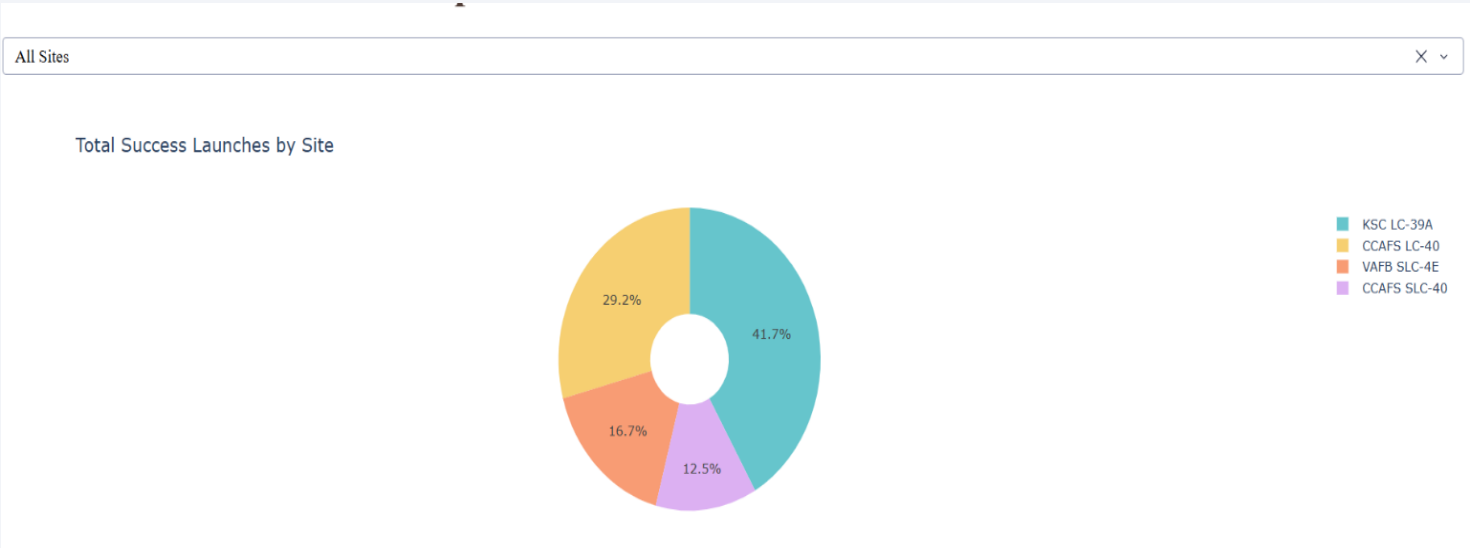
Section 4

Build a Dashboard with Plotly Dash

Aggregate Launch Success Distribution by Launch Site

1.Important Elements

- **Donut Visualization:** Represents the distribution of successful launches across all SpaceX launch sites.
- **Dynamic Filtering:** The chart is linked to the site-dropdown, allowing the viewer to toggle between "All Sites" and individual facility performance.
- **Proportional Representation:** Slice size indicates the total volume of successful missions per site, providing an immediate snapshot of operational activity.



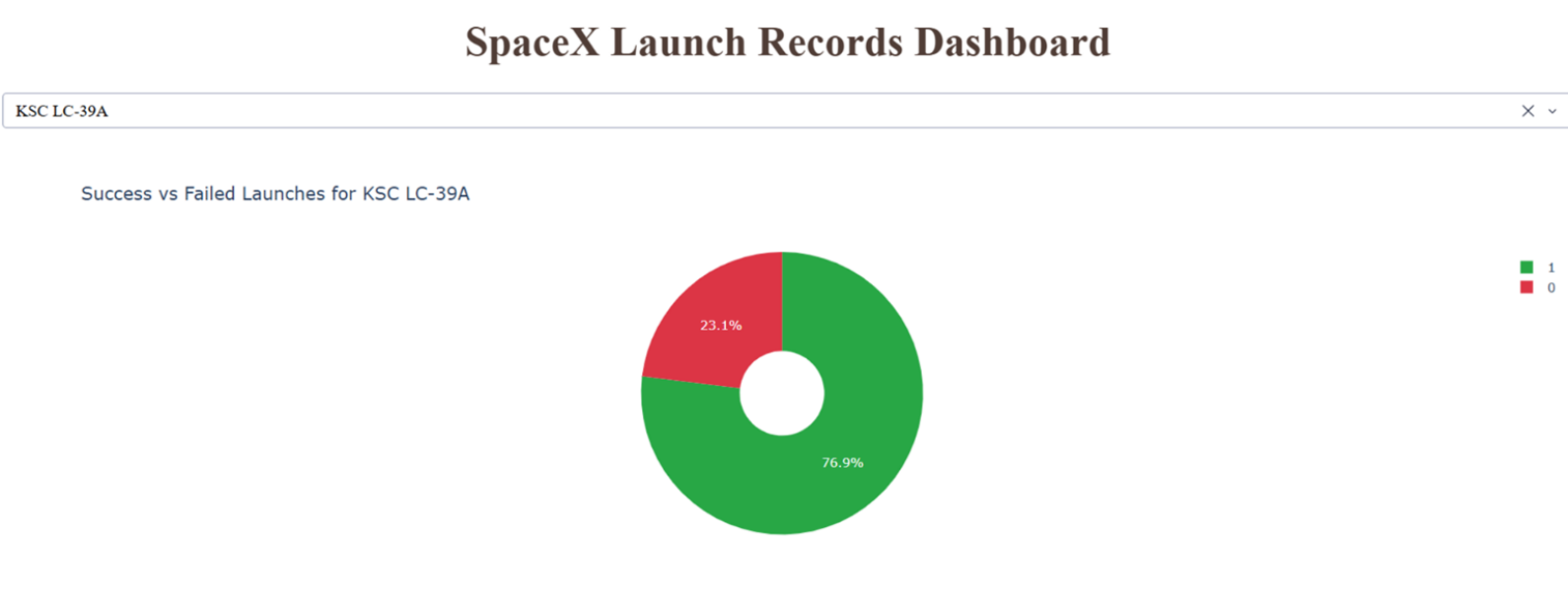
2. Key Findings

- **Facility Specialization:** The data confirms that **KSC LC-39A** is the primary hub for successful missions, indicating higher infrastructure capacity or prioritization for high-priority payloads.
- **Operational Maturity:** By comparing slice sizes, we can visualize the workload distribution across launch pads. Larger slices represent sites with established, mature launch cadences.
- **Data-Driven Decision Support:** This visualization allows stakeholders to identify "bottleneck" sites versus "high-throughput" sites, enabling better mission scheduling and launch window planning.

Launch Outcome Distribution for Highest Performing Site:KSC LC-39A

Key Findings

- **High-Performance Baseline:** The site exhibits a 76.9 % success-to-failure ratio, identifying it as the current "gold standard" for Falcon 9 mission reliability.
- **Process Maturity:** The concentration of successes validates the efficacy of existing launch protocols, ground support infrastructure, and weather risk management at this location.
- **Predictability Advantage:** High success reliability at this facility allows for lower risk-weighting in mission planning and cost-estimation models.



Payload Mass vs. Launch Success Correlation Analysis

1. Key Findings

- **Booster Evolution:** Newer booster versions show higher reliability across the payload spectrum, visually confirming engineering improvements over time.
- **Success Constraints:** The scatter plot identifies an operational "envelope",missions exceeding the max payload mass show increased outcome variance, signaling the need for high-tier booster selection for heavy lifts.

2. Important Elements

- **X-Axis (Payload Mass):** Identifies the technical load of the mission.
- **Y-Axis (Class 0/1):** Represents the binary landing outcome (0=Fail, 1=Success).
- **Color Encoding:** Groups data points by **Booster Version**, allowing for historical performance comparison.
- **Interactive Slider:** Used to isolate payload ranges and filter noise for specific mission profiles

Section 5

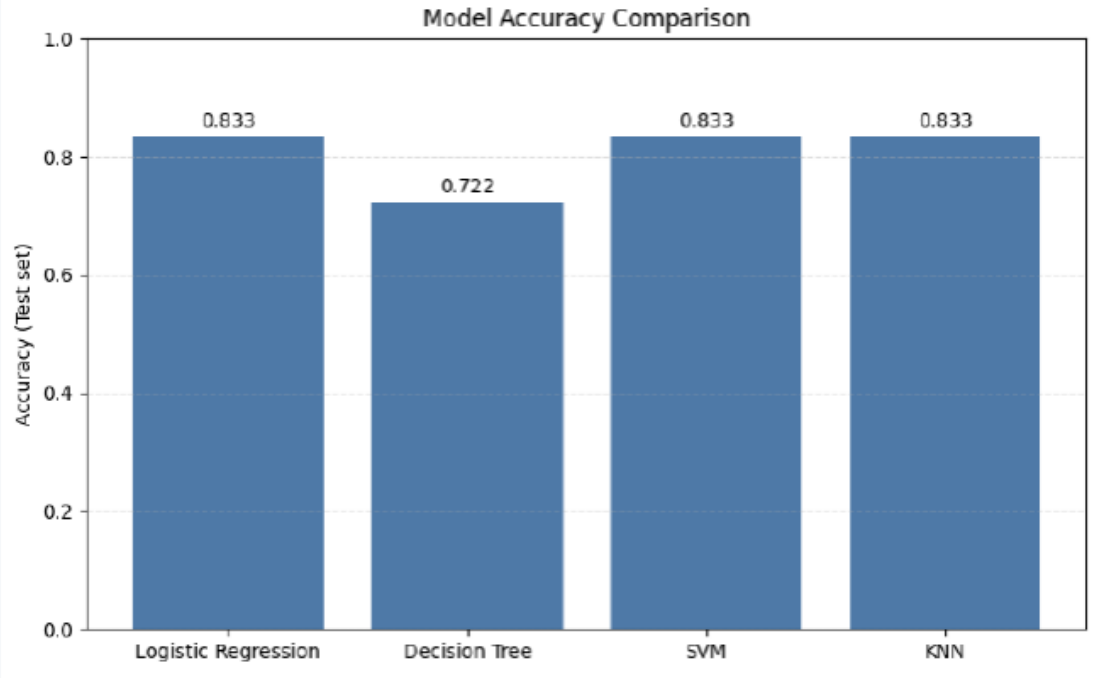
Predictive Analysis (Classification)

Classification accuracy

Model	Test Accuracy	ROC-AUC	Analysis
SVM	83%	Highest	Recommended Model
Logistic Reg.	83%	Moderate	Baseline performance.
KNN	83%	Moderate	Comparable to Logistic.
Decision Tree	< 83%	Lowest	Prone to overfitting.

Findings:

- **The Accuracy Tie:** While Logistic Regression, SVM, and KNN all achieve identical test accuracy (~83%), accuracy alone is an insufficient metric for mission-critical predictions.
- **The Overfitting Problem:** The Decision Tree demonstrates higher training/CV scores, but fails to maintain that performance on unseen test data. This indicates **overfitting**, making it less reliable for future mission predictions.
- **The SVM Advantage:** SVM is identified as the optimal model. Its superior **ROC-AUC score** confirms that it possesses the highest "discriminatory power" it is the most reliable at distinguishing between successful and failed landing profiles, even when simple classification accuracy remains tied.

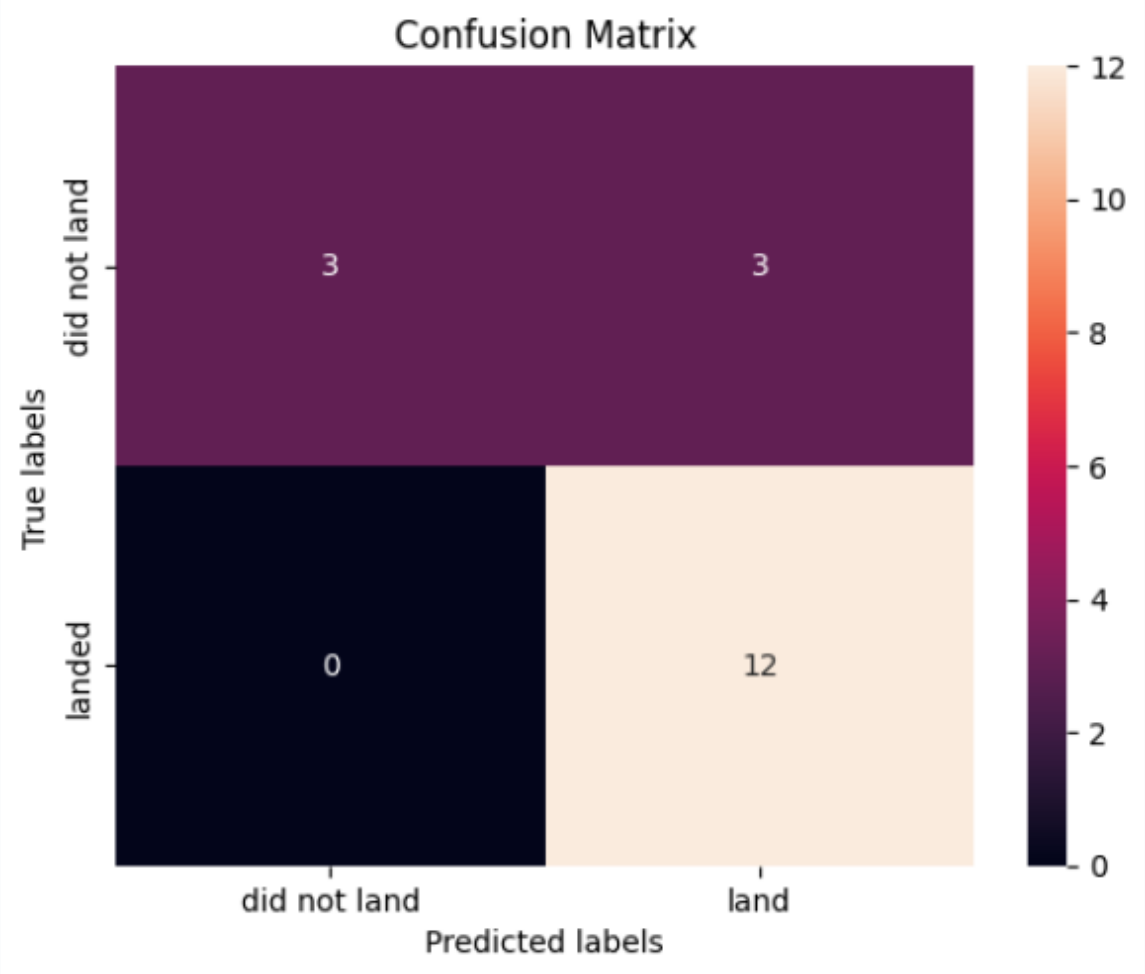


model	cv_best_mean_score	test_accuracy	ROC score:
Logistic Regression	0.846429	0.833333	0.888889
SVM	0.848214	0.833333	0.958333
KNN	0.848214	0.833333	0.895833
Decision Tree	0.878571	0.722222	0.680556

Confusion matrix

Important Elements

- **Model Performance:** Logistic Regression, SVM, and KNN demonstrated high reliability, capturing all 12 actual successful landings (True Positives).
- **The Error Analysis:** These models produced 3 False Positives. This indicates that the models are "optimistic"they have a slight tendency to predict a successful landing more often than it occurs.
- **The Decision Tree Outlier:** The Decision Tree displayed a distinct, less efficient matrix, confirming that while it is highly interpretable, it lacks the generalization capabilities of the other models.



Conclusions

Key Findings

- Falcon 9 landing success is highly predictable, not random
- Success rates vary by launch site, orbit type, payload, and time
- Clear improvement trend over time, driven by technological advancement and learning effects
- Geographic and operational factors play a critical role in mission outcomes

Model Outcome

- Support Vector Machine (SVM) identified as the **best-performing model**
- Achieved **ROC-AUC: 0.9583**, indicating strong classification capability
- Confirms the presence of non-linear relationships between mission variables and landing success

Business & Strategic Implications

- Enables more accurate cost forecasting through prediction of reusability
- Supports data-driven launch planning and optimisation
- Provides competitors with a framework to reduce uncertainty in pricing and bidding strategies

Final Perspective

"Landing success is no longer just an engineering outcome, it is a measurable, predictable phenomenon that drives SpaceX's cost-leadership strategy."

Appendix

- The links used to retrieve data from online sources:
- • SpaceX API
- https://api.spacexdata.com/v4/rockets/
- https://api.spacexdata.com/v4/launchpads/
- https://api.spacexdata.com/v4/payloads/
- https://api.spacexdata.com/v4/cores/
- https://api.spacexdata.com/v4/launches/past
- • Wikipedia
- https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_He
- avy_launches&oldid=1027686922

Karina Github Reference : https://github.com/KarinaMurugan/Karina-Murugan-IBM-Project/tree/main

Thank you!